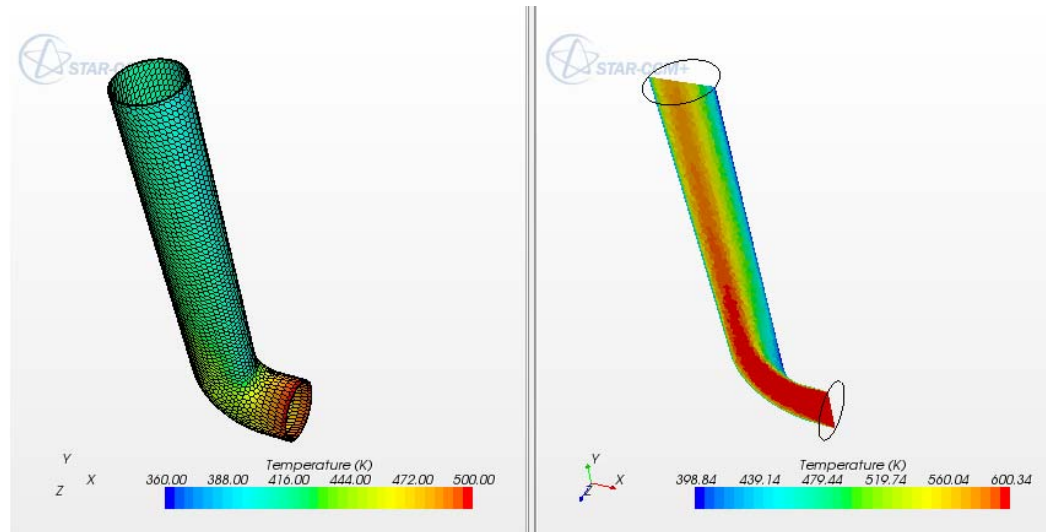




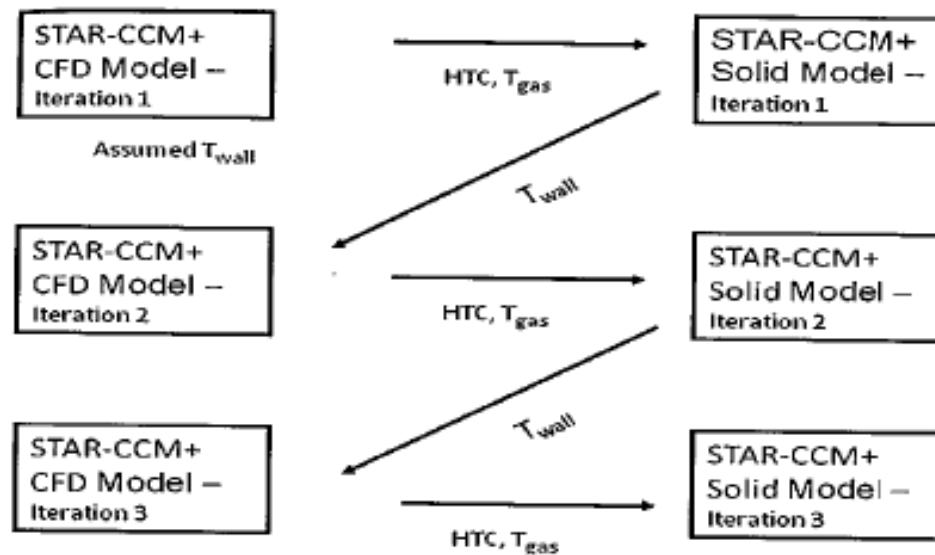
Heat Transfer Workshop (2014/11/14)



WorkShop5 STAR-CCM+弱耦合解析

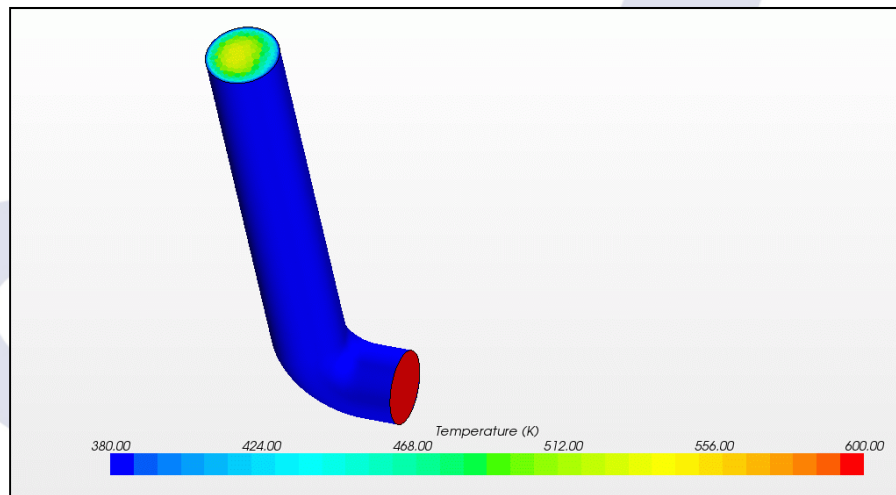
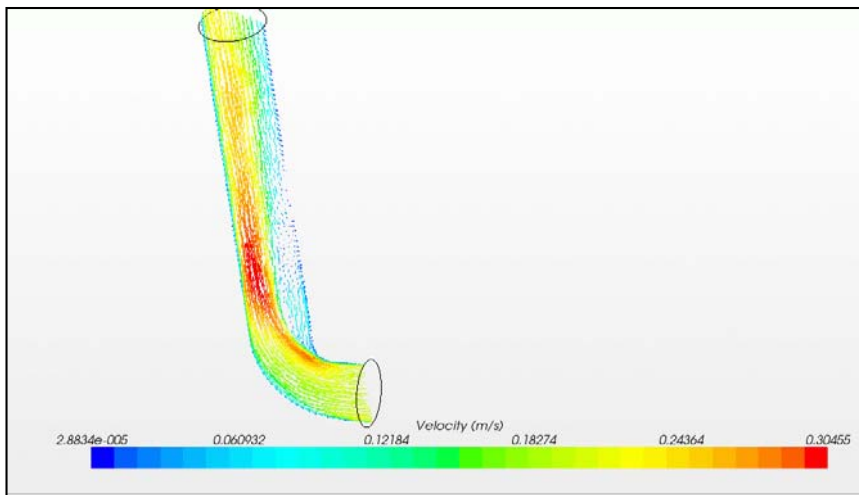
STAR-CCM+弱耦合解析

- 如果想进行流固耦合传热计算(特别是想看到在固体内温度的变化、固体计算与流体非定常计算的时间步长不在一个数量级, 这种情况下可以使用弱耦合分析。
 - 一般弱耦合, 流体使用定常解析, 固体实现非定常解析。



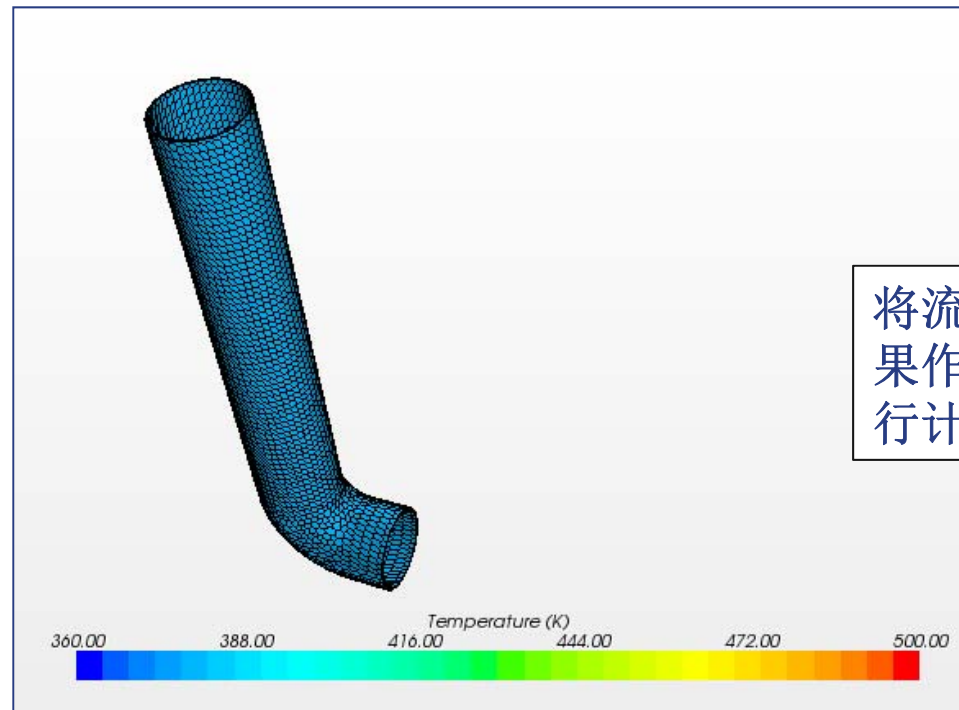
STAR-CCM+弱耦合解析

- 流体部分的定常解析结果
 - 流入壁面0.2[m/s],600[K]
 - 外侧壁面温度固定为380[K]



STAR-CCM+弱耦合解析

- 固体部分的非定常计算结果
 - 内壁温度380[K]
 - 外壁面导热系数0.12[W/m²K]、周围温度283[K]



将流体与固体已有的计算结果作为初始值，两者耦合进行计算。

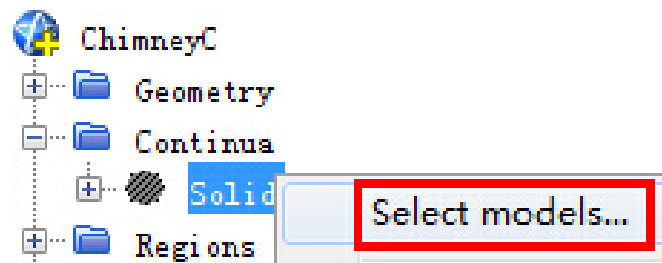
比较 Δt

- 基于Courant数Fourier数的 Δt 比较

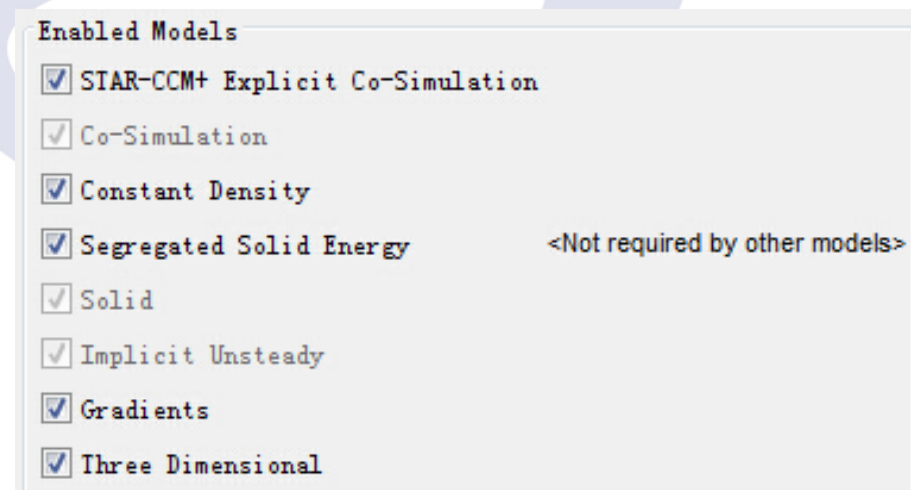
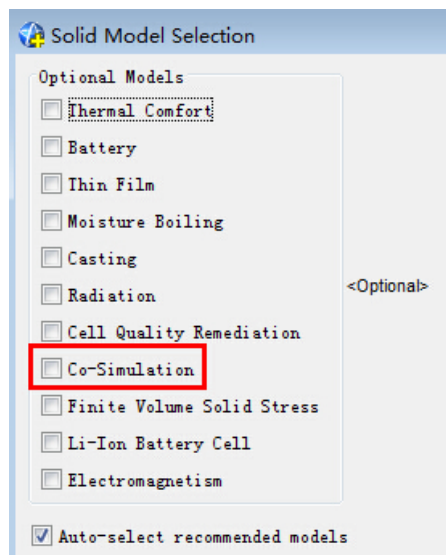
网格尺寸	0.05
热传导率	16
比热	502
密度	8000
热扩散率	3.98406E-06
固体的 Δt	627.5
有效速度	0.2
Courant数	1
流体的 Δt	0.25

选择Co-Simulation

1. File > Load Simulation > ChimneyC.sim
2. 右击[Continua] > [Solid], 选择[Select models]

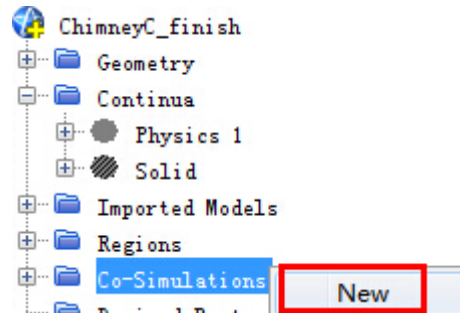


3. 在已有的物理模型中增加Co-Simulation。

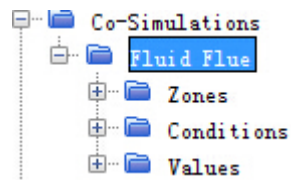


指定Concurrency Mode

1. 右击[Co-Simulations],选择 [New]

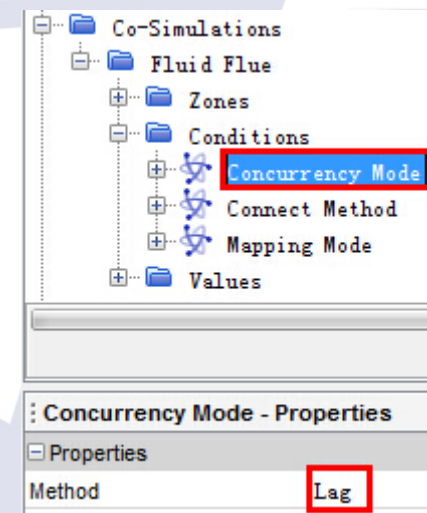


2. 重命名其为Fluid Flue



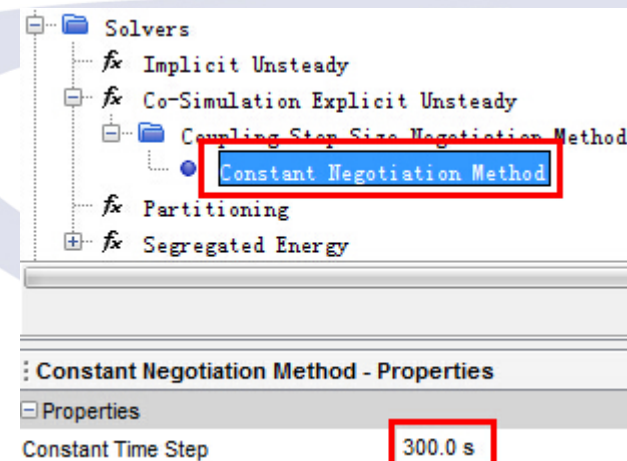
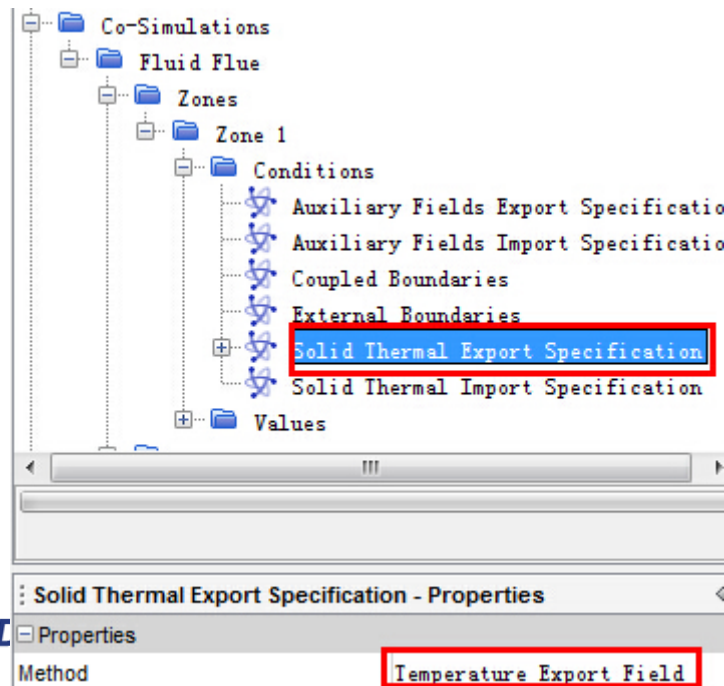
3. 在[Conditions] > [Concurrency Mode]选项中，选择 [Lag]

- 固体的计算在流体计算后进行。



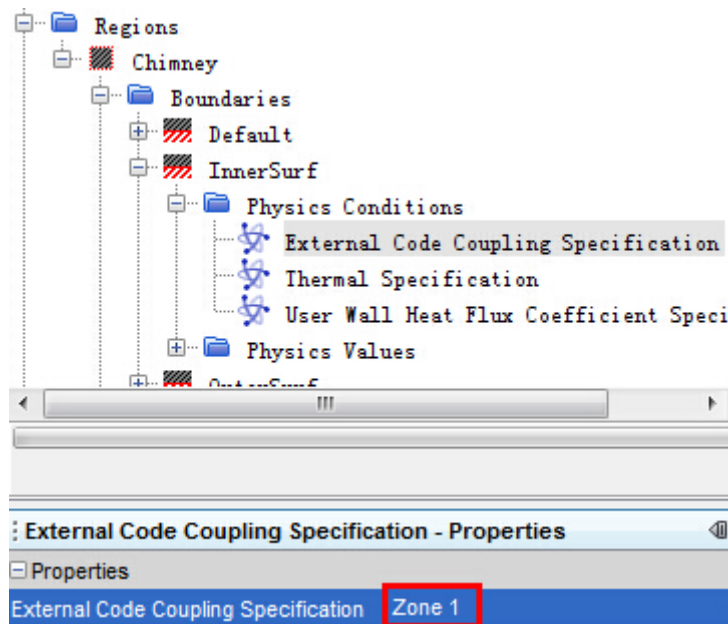
设定Co-Simulations

1. 在[Zone 1] > [Conditions] > [Solid Thermal Export Specification]中，改变其Method为Temperature Export Field
2. 在[Solvers] > [Co-Simulation Explicit Unsteady] > [Coupling step Size Negotiation Method] > [Constant Negotiation Method]中，指定Constant time step为300s
 - 在固体计算300s后，将温度传递给流体。



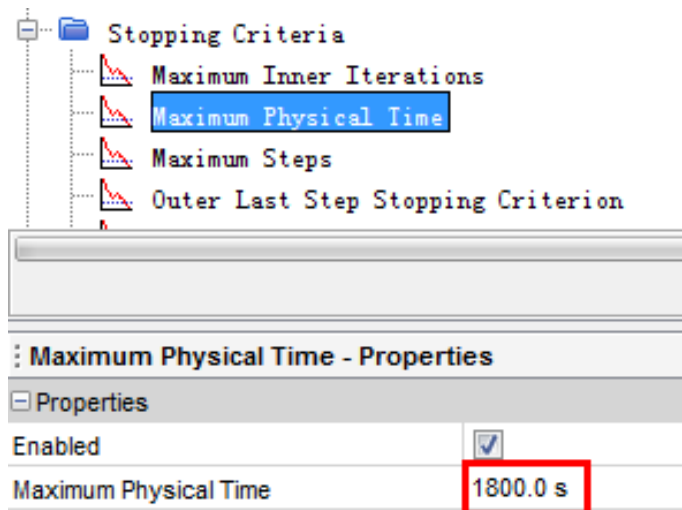
指定温度传递的边界

1. 在[Regions] > [Chimney] > [InnerSurf] > [External Code Coupling Specification]中，选择“Zone 1”。
 - 流体从固体的内侧壁面流过、从流体侧获取热传导系数。



指定计算时间

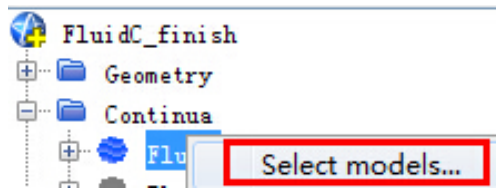
1. 设定Stopping Criteria > Maximum Physical Time为1800[s]。



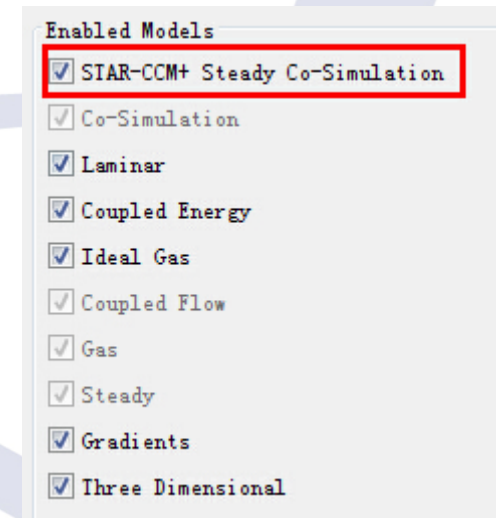
- 至此固体部分的操作结束。

读入流体模型及设定Co-Simulation

1. File > Load Simulation > FluidC.sim
2. 右击[Cintinua] > [Fluid], 选择[Select models]

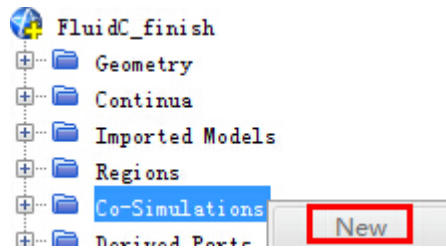


3. 在已有的模型中追加” STAR-CCM+ Steady Co-Simulation “。

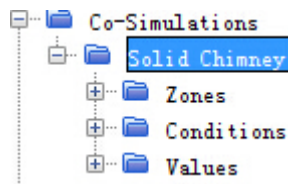


指定Concurrency Mode

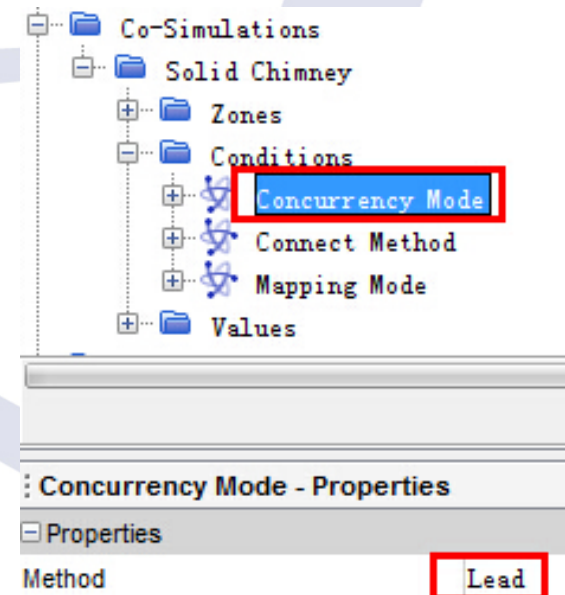
1. 右击[Co-simulations],选择[New]



2. 重命名其为Solid Chimney

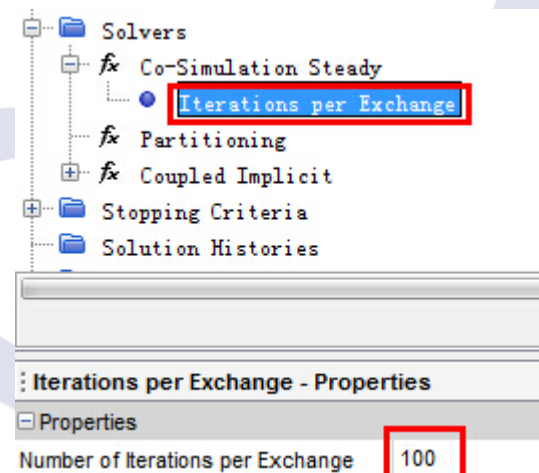
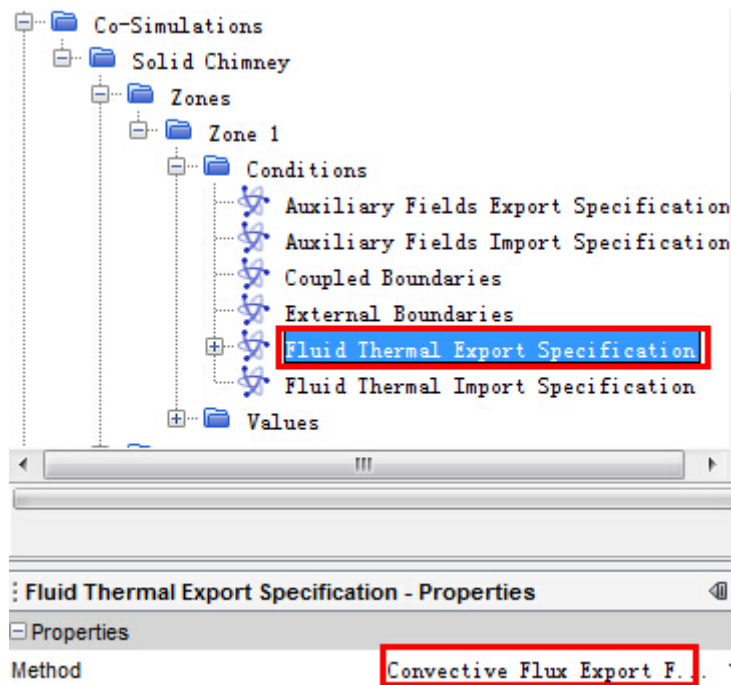


3. 在[Conditions] > [Concurrency Mode]选项中,
选择[Lead]
 - 先进行流体的计算。



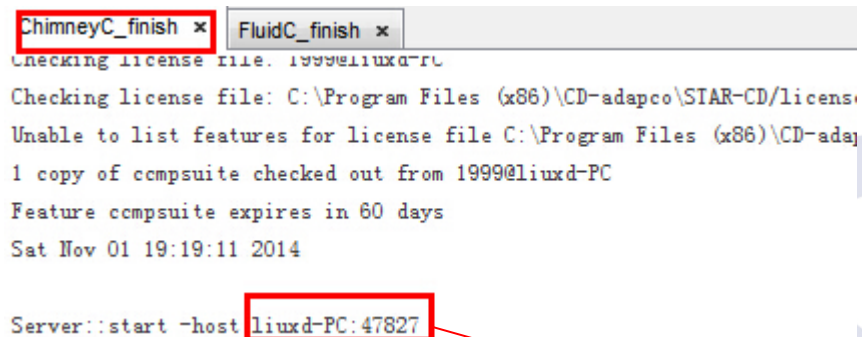
设定Co-Simulations

1. 在[Zone 1] > [Conditions] > [Fluid Thermal Export Specification]中，改变其Method为Convective Flux Export Fields
2. 设定[Solvers] > [Co-Simulation Steady] > [Iterations per Exchange]的值为100
 - 在100次迭代后将热传导系数传递给固体。



设定Co-Simulations

1. Output窗口 > ChimneyC
复制机器名以及端口号
2. 在[Conditions] > Connect Method中，选择Connect to Partner Simulation，在Connect to STAR-CCM+ Server Process中填入刚才复制的机器名以及端口号。



ChimneyC_finish x FluidC_finish x

Checking license file: 1999@liuxd-PC

Checking license file: C:\Program Files (x86)\CD-adapco\STAR-CD\licens

Unable to list features for license file C:\Program Files (x86)\CD-adap

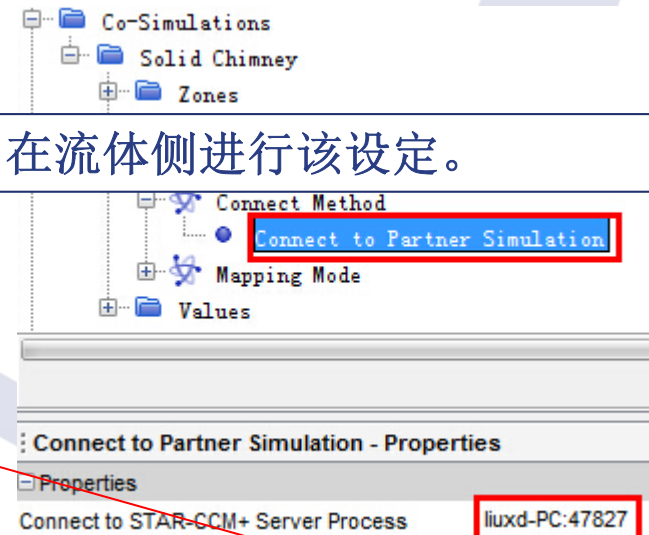
1 copy of ccpsuite checked out from 1999@liuxd-PC

Feature ccpsuite expires in 60 days

Sat Nov 01 19:19:11 2014

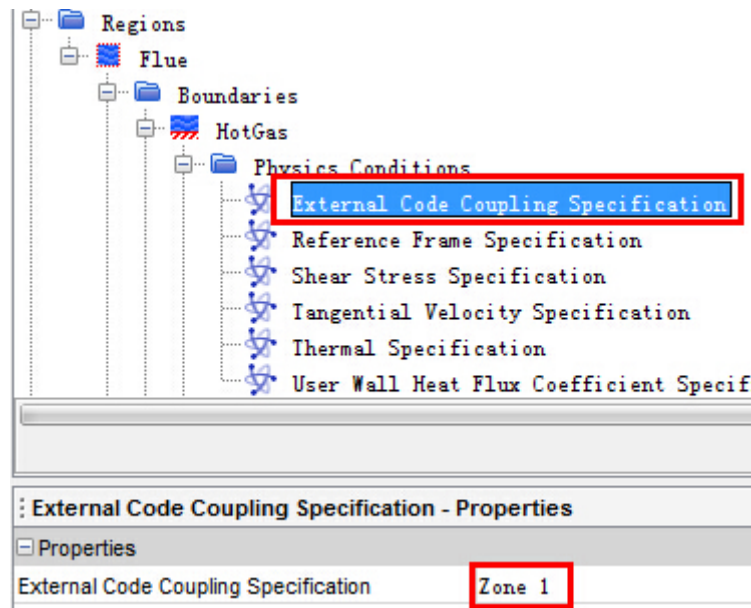
Server::start -host liuxd-PC:47827

在流体侧进行该设定。



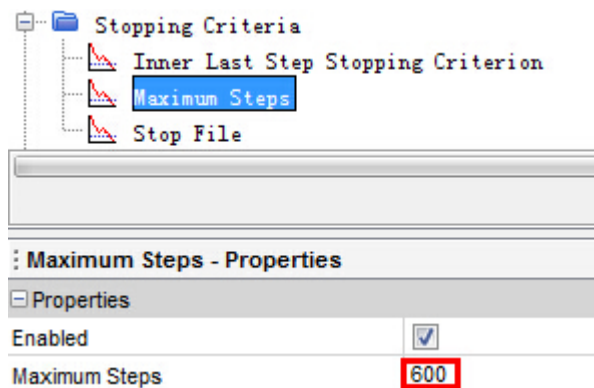
指定温度传递的边界

1. 在[Regions] > [Flue] > [HotGas] > [External Code Coupling Specification]中，选择“Zone 1”。



实行计算

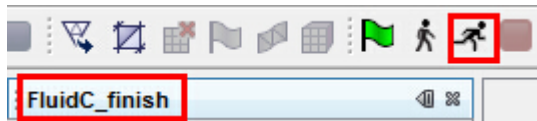
1. 将[Stopping Criteria] > [Maximum Steps]改为600。



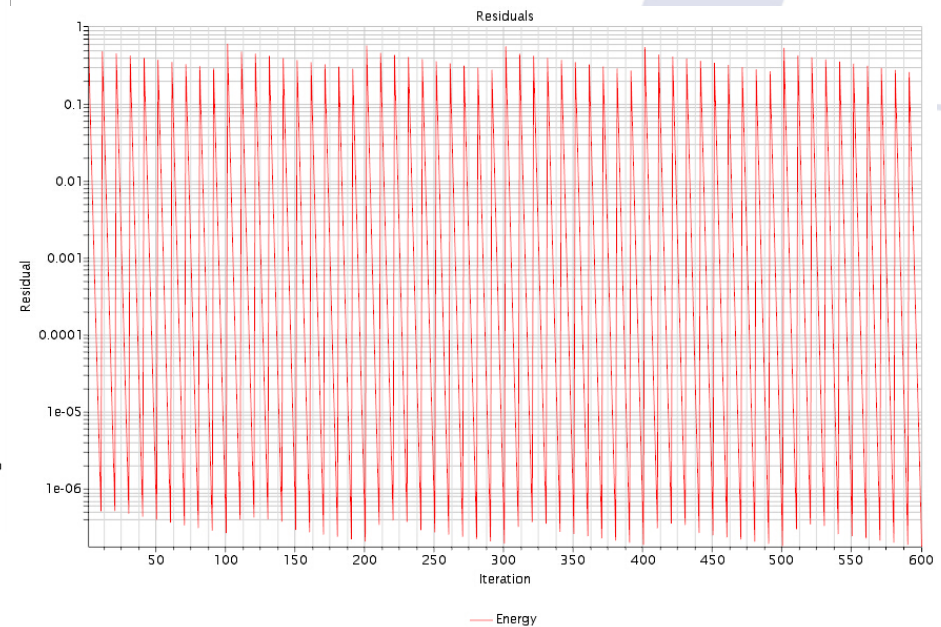
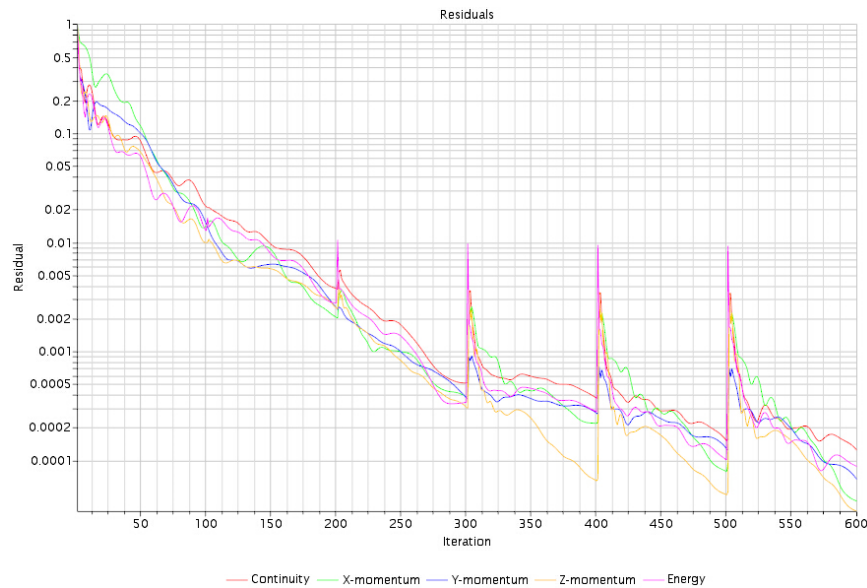
2. 保存sim文件

实行计算

1. 从流体侧开始计算。



- 流体计算开始后，固体计算将会自动开始。



计算结果

1. 打开已有的视图

- Chimney C的[Wall Temperature]视图
- FluidC的[Fluid Temperature]视图

